BioOne DIGITAL LIBRARY

INVESTIGATING PUTATIVE HYBRIDS OF FREMONTODENDRON DECUMBENS AND F. CALIFORNICUM IN THE SIERRA FOOTHILLS

Authors: Kenny, Reed J., McMahan, Will T., Ayres, Debora, Meyer, Virginia C., Grotkopp, Eva, et al.

Source: Madroño, 72(3) : 111-125

Published By: California Botanical Society

URL: https://doi.org/10.3120/0024-9637-250014

The BioOne Digital Library (https://bioone.org/) provides worldwide distribution for more than 580 journals and eBooks from BioOne's community of over 150 nonprofit societies, research institutions, and university presses in the biological, ecological, and environmental sciences. The BioOne Digital Library encompasses the flagship aggregation BioOne Complete (https://bioone.org/subscribe), the BioOne Complete Archive (https://bioone.org/archive), and the BioOne eBooks program offerings ESA eBook Collection (https://bioone.org/esa-ebooks) and CSIRO Publishing BioSelect Collection (https://bioone.org/csiro-ebooks).

Your use of this PDF, the BioOne Digital Library, and all posted and associated content indicates your acceptance of BioOne's Terms of Use, available at www.bioone.org/terms-of-use.

Usage of BioOne Digital Library content is strictly limited to personal, educational, and non-commmercial use. Commercial inquiries or rights and permissions requests should be directed to the individual publisher as copyright holder.

BioOne is an innovative nonprofit that sees sustainable scholarly publishing as an inherently collaborative enterprise connecting authors, nonprofit publishers, academic institutions, research libraries, and research funders in the common goal of maximizing access to critical research.

Downloaded From: https://bioone.org/journals/Madroño on 17 Feb 2026
Terms of Use: https://bioone.org/terms-of-use

MADROÑO, Vol. 72, No. 3, pp. 111–125, 2025
DOI:10.3120/0024-9637-250014

INVESTIGATING PUTATIVE HYBRIDS OF *FREMONTODENDRON DECUMBENS* AND *F. CALIFORNICUM* IN THE SIERRA FOOTHILLS

REED J. KENNY
Graduate Group in Ecology and Department of Plant Sciences, University of California, Davis, One Shields Avenue, Davis, CA 95616
reedjohnkenny@gmail.com

WILL T. MCMAHAN
Graduate Group in Horticulture and Agronomy and Department of Plant Sciences, University of California, Davis, One Shields Avenue, Davis, CA 95616

DEBORA AYRES
Department of Evolution and Ecology, University of California, Davis, One Shields Avenue, Davis, CA 95616

VIRGINIA C. MEYER
Professor Emerita, Sacramento City College, 3835 Freeport Boulevard, Sacramento, CA 95822

EVA GROTKOPP
Department of Plant Biology, University of California, Davis, One Shields Ave, Davis, CA 95616

SHANNON M. STILL
H. T. Harvey and Associates, 255 Parkshore Drive, Suite 150, Folsom, CA 95630

DANIEL POTTER
Department of Plant Sciences, University of California, Davis, One Shields Avenue, Davis, CA 95616

ABSTRACT

Fremontodendron (Flannelbush, Fremontia, Malvaceae) is a genus of shrubs native to California, Arizona, and Baja California Norte. There are three species in the genus: *Fremontodendron californicum* (Torr.) Coville is widespread throughout the range and is highly variable in its appearance, *F. mexicanum* Davidson is endemic to Otay Mountain in southern California, and *F. decumbens* R.M.Lloyd is endemic to gabbro soils in the Sierra Nevada foothills. *Fremontodendron decumbens* is currently known only from the Pine Hill Preserve and adjacent areas in El Dorado County and was listed as endangered in 1996. *Fremontodendron decumbens* is mainly distinguished from *F. californicum* by its orange or reddish flowers and decumbent habit. Many plants with a decumbent habit but lacking the floral characters to be placed in *F. decumbens* have been collected over the years, primarily in the Northern Sierra Nevada/Cascade foothills, and there has long been speculation as to whether these could be hybrids between *F. californicum* and *F. decumbens*. Here we investigated the potential hybrid origin of morphologically intermediate individuals using full genome resequencing data from 99 individuals of *F. californicum*, *F. decumbens*, and potential hybrids, mapped to a *F. californicum* reference genome. We investigated population structure and potential admixture using principal components analysis, model-based evolutionary clustering, and D-statistics, as well as additional analyses. We also systematically investigated the morphology of *F. californicum*, *F. decumbens*, and potentially intermediate individuals by measuring a suite of ten characters in the field on 68 individuals. We found no evidence that the morphologically intermediate forms are of hybrid origin; rather, they generally group genetically with *F. californicum*. We also found that specimens from populations noted to be morphologically intermediate showed considerable overlap with *F. californicum*, but not *F. decumbens*, based on a morphological principal components analysis. The results lead us to conclude that *F. decumbens* is indeed a genetically distinct taxon and that individuals of *F. californicum* with a decumbent habit, but lacking the distinct floral characters of *F. decumbens*, are not the result of hybridization.

Key Words: Endangered species, *Fremontodendron*, hybridization, population genetics.

Downloaded From: https://bioone.org/journals/Madroño on 17 Feb 2026
Terms of Use: https://bioone.org/terms-of-use

Fremontodendron (Flannel bush, Fremontia) is a genus of shrubs in the Malvaceae (Mallow family) native throughout much of California and parts of Arizona and Baja California Norte. There are three species in the genus, all having characteristic densely stellate-hairy leaves and large apetalous flowers with yellow to orange (red) sepals (Kelman 1991). *Fremontodendron californicum* (Torr.) Coville is widespread throughout the range and is highly variable in its appearance. It is a distinctive and common member of chaparral communities in the Sierra Nevada foothills as well as the Coast Ranges. *Fremontodendron mexicanum* Davidson is endemic to Southern California, found only on one mountain in the Otay wilderness, and was listed as federally endangered in 1998 (USFWS 2002). *Fremontodendron decumbens* R.M.Lloyd is endemic to gabbro soils in the Sierra Nevada foothills; it is currently known only from several occurrences on and around the Pine Hill Preserve in El Dorado County and was listed as federally endangered in 1996 (USFWS 1996).

Fremontodendron decumbens was described from a specimen collected on the Pine Hill formation by Beecher Crampton in 1956. She noted that it was a "sprawling, semi-prostrate shrub" with red-orange to reddish yellow flowers (Lloyd 1965). The Pine Hill formation is an intrusive complex of gabbro parent material (Medeiros et al. 2015). Gabbro is a plutonic igneous rock formed by the subterranean solidification of mafic magma. It can have a range of chemical properties and often weathers into silty or clayey soils that inherit these properties. Depending on site-specific factors, gabbro soils may be limiting in potassium, phosphorus, calcium, or magnesium but are unlikely to have metal toxicity (Medeiros et al. 2015).

Fremontodendron decumbens is distinguished from *F. californicum* by its orange or reddish flowers and decumbent habit (Fig. 1) (Kelman 1991). Many plants with a decumbent habit but lacking the floral characters to be placed in *F. decumbens* have been collected over the years, primarily in the Northern Sierra Nevada/Cascade foothills. These have primarily come from Yuba, Butte, and Nevada counties (Kelman et al. 2006). Because these individuals are morphologically intermediate, there has long been speculation as to whether these could be hybrids between *F. californicum* and *F. decumbens*, whether they should be appropriately treated as *F. decumbens*, or whether they could be a new species (Kelman et al. 2006). Here they are referred to as *F.* aff. *decumbens*.

The recovery plan for Gabbro Soil Plants of the Sierra Nevada Foothills (2002) recommended that "the decumbent *Fremontodendron* within Yuba and Nevada counties should be secured and protected unless they are determined not to be the listed [*F. decumbens*]" (USFWS 2002). As all of the known populations of *F. decumbens* occur on or close to the Pine Hill preserve, identifying new populations would represent a significant step towards meeting

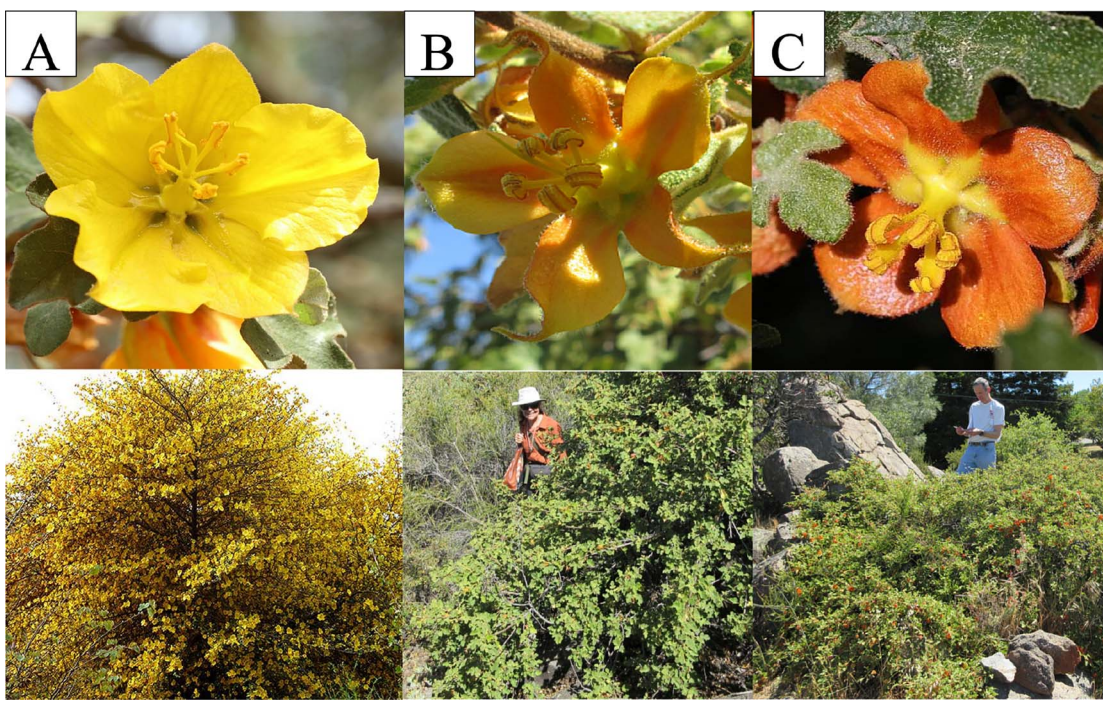

FIG. 1. *Fremontodendron californicum* (A), *F.* aff. *decumbens* (B), *F. decumbens* (C). Top photos show the floral color, bottom shows the habit. Photos provided by Debra Ayres.

Downloaded From: https://bioone.org/journals/Madroño on 17 Feb 2026
Terms of Use: https://bioone.org/terms-of-use

the goals of the recovery plan. Conversely, conclusively determining that these plants are not *F. decumbens* or recent hybrids would allow managers to have a more accurate assessment of the population status of *F. decumbens* and focus resources on protecting and expanding the Pine Hill populations. If these populations were to represent a new species, it is likely that it would be rare and deserve conservation recognition. Finally, if these populations were to be identified as recent or ancestral hybrids of *F. decumbens*, there could be a variety of conservation implications as well as potential further studies. The conservation value of hybrid individuals is a complex question. In some circumstances, ongoing hybridization of a rare species with a common or introduced species is a main threat to the persistence of the rare species (Ryan et al. 2009). In other cases, hybrid populations may have been formed at discrete time periods and persist as a separate population, perhaps in similar ecological circumstances, and represent a continuance of a rare genetic heritage (Gross and Rieseberg 2005).

A previous study of the morphology and genetics, based on AFLP data, of the decumbent *Fremontodendron* in Yuba and Nevada counties found that they did not group conclusively with either *F. decumbens* or *F. californicum* (Kelman et al. 2006). However, based on the presence of unique alleles in the decumbent populations they genotyped, they suggested continued conservation until further data become available. This study speculated that the decumbent populations in Yuba and Nevada counties are likely to be the product of hybridization between *F. decumbens* and *F. californicum*.

Given the importance of determining the genetic status of the decumbent *Fremontodendron* found outside of the Pine Hill area (hereafter referred to as *F.* aff. *decumbens*), we tested the hypothesis that *F.* aff. *decumbens* populations are derived from the hybridization of *F. californicum* and *F. decumbens*, either from recent hybridization events where there has been minimal backcrossing with *F. californicum*, or as a result of ancestral hybridization events with low rates of backcrossing to the parental genotypes. If this hypothesis is true, we expect to see either genetic intermediacy of the *F.* aff. *decumbens* individuals, which would support recent hybridization, or the presence of alleles associated with *F. decumbens*, as can be detected using a variety of population genetics tests.

To test this hypothesis, we generated full genome sequences for 99 individuals of *F. californicum*, *F. decumbens*, and *F.* aff. *decumbens*, in collaboration with the California Conservation Genomics Project, a state-funded effort to produce genomic data to manage regional biodiversity (https://www.ccgproject.org/about-us). We used PCA, ADMIXTURE, and outgroup f3 analyses to assess the genetic intermediacy of the *F.* aff. *decumbens* individuals and Treemix and D statistics to test for the presence of alleles associated with *F. decumbens* in the *F.* aff. decumbens individuals. We also measured morphological traits on 68 individuals in the field to characterize the degree of morphological intermediacy seen in *F.* aff. *decumbens* individuals.

METHODS

Leaf Sample Collection

We collected leaf samples from 99 individuals across 12 sites in Nevada, Yuba, El Dorado, Napa, Santa Cruz, and Tehama counties (Fig. 2). We collected from between 1 and 19 individuals per site (mean 13 individuals) and immediately stored leaves on silica gel. Herbarium voucher specimens were collected for most populations, and GPS locations were recorded for each collection site (Appendix S1).

DNA Extraction, Library Preparation, and Sequencing

For samples prepared at UC Davis, total genomic DNA was extracted using 20 mg or less of lyophilized leaf tissue frozen in liquid nitrogen and ground to a fine powder using 3 mm ceramic balls in a BeadBug (BeadBug 3 Microtube homogenizer model D1030, Benchmark Scientific, Sayreville, NJ) bead mill and extracted using a modified CTAB protocol with an additional cleaning step using the DNeasy PowerClean Pro Cleanup kit (Qiagen, Hilden, Germany). Whole genome libraries were prepared following the plexWell library preparation protocol by the UC Davis Genome Center and paired-end sequencing (2×150 bp) was performed on an Illumina HiSeq4000 platform at the UC Davis Genome Center (Davis, CA).

For samples prepared by the CCGP, genomic DNA was extracted from leaf tissue using NucleoMag Bacteria DNA Isolation kit (Macherey-Nagel, Düren, Germany), and whole-genome libraries were prepared following the plexWell library preparation protocol by the CCGP MiniCore. Paired-end sequencing (2×150 bp) was performed on an Illumina HiSeq4000 platform at the UCLA MiniCore facility (Los Angeles, CA). Raw reads from both sequencing efforts can be downloaded from GenBank BioProject (PRJNA1041826).

Genetic Analyses

Processing raw reads. We processed all raw reads together using the SNParcher pipeline, available at (https://github.com/harvardinformatics/snpArcher), with the default settings (Mirchandani et al. 2024). Sequencing adapters were trimmed from the raw sequence files and sequences with greater than 40% of bases with a phred score below Q15 were removed using *fastp* (Chen et al. 2018). Sequencing reads were aligned to the *Fremontodendron californicum* reference genome (GenBank GCA_030761205.1) with a

Downloaded From: https://bioone.org/journals/Madroño on 17 Feb 2026
Terms of Use: https://bioone.org/terms-of-use

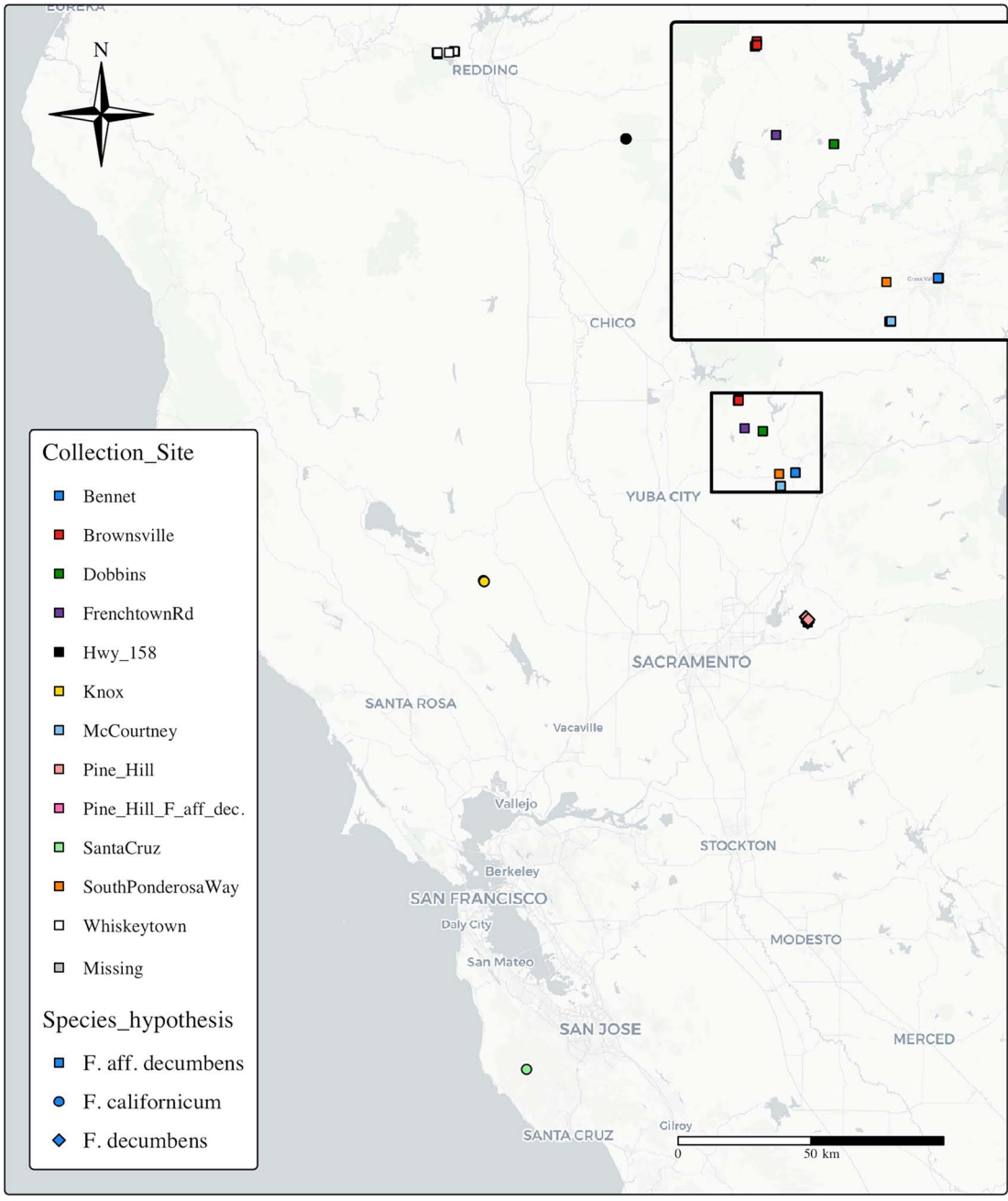

FIG. 2. Map of all *Fremontodendron* individuals sampled. Point colors represent collection site, while point shapes represent the species hypothesis based on morphology. Inset shows zoomed-in view of sites in Yuba and Nevada counties.

Fremontodendron californicum chloroplast genome added as an additional chromosome using the Burrows-Wheeler aligner (BWA-MEM) algorithm (Li and Durbin 2009). PCR duplicates were marked using *sambamba markdup* and excluded from downstream analysis (Tarasov et al. 2015). Joint genotyping was performed on all samples using Genome Analysis Toolkit (GATK4) version 4.1.8 functions HaplotypeCaller and GenotypeGVCFs (Poplin et al. 2017). Variant data were filtered to include only biallelic single nucleotide polymorphisms (SNPs), and SNPs were hard filtered using GATK best practice recommendations. SNPs were removed if they had a minor allele frequency of less than 0.01, or greater than 75% missing data, and samples were removed if they had less than 2× sequencing depth. Raw read

Downloaded From: https://bioone.org/journals/Madroño on 17 Feb 2026
Terms of Use: https://bioone.org/terms-of-use

processing was performed by the California Conservation Genomics Project bioinformatics team. As variant filtering thresholds have been shown to influence estimates of population structure, we assessed sensitivity of genetic differentiation to filter thresholds (Linck and Battey 2019).

Population genomics statistics. We calculated individual-level heterozygosity and inbreeding coefficient using VCFtools version 0.1.16 on the full dataset (Danecek et al. 2011). We also calculated the mean per site nucleotide diversity (π), and Wier and Cockerham Fst, grouped by taxonomic assignment (*F. californicum*, *F. decumbens*, and *F.* aff. *decumbens*) as well as by collection site, using pixy version 1.2.10 (Korunes and Samuk 2021). To do this we first generated VCF files that included invariant sites from the GVCF files output by the SNParcher pipeline using the gatk version 4.2.6.1 functions DBimport and GenotypeGVCFs (Auwera and O'Connor 2020). We then used pixy to calculate nucleotide diversity and Fst for each chromosome independently using a sliding window of 100,000 bps. We calculated the average nucleotide diversity for each group by summing the count of pairwise differences between all genotypes in each window and the count of all non-missing comparisons between all genotypes in the window, then dividing the first count by the second.

Genomic principal components analysis. We performed an initial investigation of population structure using principal components analysis (PCA). First, we pruned our VCF file to remove any sites in linkage disequilibrium (LD) ($r^2 > 0.1$) using plink version 1.9 (https://www.cog-genomics.org/plink/) with a sliding window of 50 kb and a step size of 10 base pairs. We then calculated a genomic PCA using plink (Purcell et al. 2007). PCA was also performed using the 122 SNPs derived from the chloroplast data; however, no groups were separated. Because of the lack of differentiation, we do not include any further analysis on the chloroplast data alone in our results. The chloroplast PCA as well as a maximum likelihood phylogeny generated from chloroplast SNPs can be found in Appendix S2 Figs. 1 and 2.

ADMIXTURE. We further investigated population structure as well as potential hybridization using a model-based evolutionary clustering approach as implemented in ADMIXUTRE version 1.3 (Alexander et al. 2009). We used the same LD pruning method as for the PCA analysis. We ran ADMIXTURE under K = 2, 3, and 4, representing two, three, or four ancestral populations respectively. Each K value was repeated 5 times with a different random seed, and cross-validation errors were calculated. Given that we are testing the hypothesis that *F.* aff. *decumbens* individuals are hybrids between *F. decumbens* and *F. californicum*, implicit in this hypothesis is the assumption that both *F. decumbens* and *F. californicum* represent genetically cohesive, single species. This assumption is supported by prior morphological analyses (Kelman 1991). Following this assumption, *F. decumbens* and *F. californicum* individuals should not be admixed, and we can treat them as reference individuals. To take advantage of this, we ran ADMIXTURE in a supervised manner, under K = 2, where all samples identified morphologically to be either *F. californicum* or *F. decumbens* were assigned 100% ancestry from their respective species and only ancestry proportions for samples identified as *F.* aff. *decumbens* were calculated. This reduces the uncertainty in allele frequencies, the number of parameters to estimate, and can reduce the error rates in the model (Alexander and Lange 2011). To explore the full range of likely K values, we ran models assuming up to five ancestral states and calculated their cross-validation error.

F3 statistics. F3 statistics can be fit to population genetic data to ascertain to what degree the populations represented can be placed on a tree with bifurcating branches (Peter 2022). If the calculated f3 statistic comes out as a negative, this violates the assumption of a tree with bifurcating branches and branch lengths proportional to genetic drift. The alternative assumption is often that the more appropriate model is that of a phylogenetic network, where branches can be reticulate, representing hybridization. We generated a file with no missing data and removed sites in LD using the same pruning methods as above. We fit f3 statistics using the THREEPOP module contained within Treemix version 1.13 (Pickrell and Pritchard 2012), with samples assigned to populations based on their morphological identification (*F. californicum, F. decumbens, F.* aff *decumbens*). This approach calculates an f3 statistic for all possible triplets of populations in the form of (A;B, C), where a significantly negative value is consistent with A being a hybrid of B and C (Hubisz et al. 2009). We calculated standard error of the f3 statistic using blocks of 500 SNPs as recommended by the program authors. We additionally fit f3 statistics with samples assigned to groups based on geographic location. This resulted in nine groups with more than four individuals each, consisting of plants located within one km of each other. We fit f3 statistics for every possible triplet of the nine groups and calculated standard error as above.

Treemix. We used the program Treemix version 1.13 to infer a tree topology and admixture jointly, using pooled SNP frequency data (Pickrell and Pritchard 2012). We included 12 *F. mexicanum* individuals, collected at Otay Mountain and processed identically to our other samples, in order examine the effect of rooting the tree on a more genetically distant group. We pooled individuals into clades based on collection location. We excluded collection sites where we collected fewer than three individuals, as Treemix applies a sample size correction. We filtered our VCF to remove all missing data and inferred topologies without admixture as well as with up to eight migration events, with *F. mexicanum*,

Downloaded From: https://bioone.org/journals/Madroño on 17 Feb 2026
Terms of Use: https://bioone.org/terms-of-use

TABLE 1. D STATISTICS CALCULATED USING ADMIXTOOLS AND ADMIXR. Calculations are based on the topology (((W, X) Y) Z). Z scores with an absolute value greater than three are generally accepted as significant. Positive D and Z scores indicate introgression between W and Y and negative scores indicate introgression between X and Y. The last two columns list the raw numbers of SNPs supporting a BABA or ABBA topology respectively.

W	X	Y	Z	D	Z score	BABA	ABBA
F. mex.	Bennet	Pine_Hill	Hwy 158	−0.1213	−37.062	16203	20679
F. mex.	McCourtney	Pine_Hill	Hwy 158	−0.0707	−20.399	16539	19056
F. mex.	Brownsville	Pine_Hill	Hwy 158	−0.081	−20.229	15614	18367
F. mex.	Dobbins	Pine_Hill	Hwy 158	−0.0989	−22.018	14699	17926
F. mex.	FrenchtownRd	Pine_Hill	Hwy 158	−0.0827	−19.342	15309	18071
F. mex.	Bennet	Pine_Hill	Knox	−0.022	−8.499	15036	15712
F. mex.	McCourtney	Pine_Hill	Knox	0.108	40.153	16835	13553
F. mex.	Brownsville	Pine_Hill	Knox	0.0893	25.073	15315	12804
F. mex.	Dobbins	Pine_Hill	Knox	0.1052	25.606	14927	12087
F. mex.	FrenchtownRd	Pine_Hill	Knox	0.0996	25.869	15176	12427
F. mex.	Bennet	Pine_Hill	F. mex.	−5.00E-04	−0.194	15695	15712
F. mex.	McCourtney	Pine_Hill	F. mex.	0.0312	13.236	14427	13553
F. mex.	Brownsville	Pine_Hill	F. mex.	0.0257	8.842	13478	12804
F. mex.	Dobbins	Pine_Hill	F. mex.	0.0176	4.878	12520	12087
F. mex.	FrenchtownRd	Pine_Hill	F. mex.	0.031	9.366	13223	12427
Hwy 158	Bennet	Pine_Hill	F. mex.	−0.1371	−40.376	15692	20679
Hwy 158	McCourtney	Pine_Hill	F. mex.	−0.1204	−34.343	14960	19056
Hwy 158	Brownsville	Pine_Hill	F. mex.	−0.1324	−34.097	14071	18367
Hwy 158	Dobbins	Pine_Hill	F. mex.	−0.1449	−35.367	13389	17926
Hwy 158	FrenchtownRd	Pine_Hill	F. mex.	−0.1306	−33.091	13896	18071
Hwy 158	Whiskeytown	Pine_Hill	F. mex.	−0.1171	−33.382	15835	20034
Knox	Bennet	F. mex.	Pine_Hill	5.00E-04	0.194	15712	15695
Knox	McCourtney	F. mex.	Pine_Hill	−0.0312	−13.236	13553	14427
Knox	Brownsville	F. mex.	Pine_Hill	−0.0257	−8.842	12804	13478
Knox	Dobbins	F. mex.	Pine_Hill	−0.0176	−4.878	12087	12520
Knox	FrenchtownRd	F. mex.	Pine_Hill	−0.031	−9.366	12427	13223
Hwy 158	Bennet	F. mex.	Pine_Hill	0.1371	40.376	20679	15692
Hwy 158	McCourtney	F. mex.	Pine_Hill	0.1204	34.343	19056	14960
Hwy 158	Brownsville	F. mex.	Pine_Hill	0.1324	34.097	18367	14071
Hwy 158	Dobbins	F. mex.	Pine_Hill	0.1449	35.367	17926	13389
Hwy 158	FrenchtownRd	F. mex.	Pine_Hill	0.1306	33.091	18071	13896

F. californicum, and *F. decumbens* specified as the roots. We used the R package *OptM* to investigate the optimum number of migration edges to add (Fitak 2021). Trees with migration edges were visualized in R using the *popcorn* package (https://github.com/andrewparkermorgan/popcorn/).

Maximum likelihood phylogeny. We constructed a maximum likelihood phylogeny of individuals using IQ-TREE (Hoang et al. 2018; Minh et al. 2020). We included the same 12 *F. mexicanum* individuals as above and used vcf2phylip v. 2 to convert our linkage pruned and no missing data VCF to fasta format (Ortiz 2019). We specified GTR + ASC as the model to account for ascertainment bias in only having variant sites in the matrix and ran IQ-TREE with 1000 ultrafast bootstraps to calculate branch support values. We visualized the resulting tree using the R package GGtree and rooted it on *F. mexicanum* for the purposes of visualization as our unpublished data indicates *F. mexicanum* to be more genetically distant from the focal taxa (Yu 2020).

D statistics. We used the R package admixr (Petr et al. 2019) and its dependency ADMIXTOOLS v1 (Patterson et al. 2012) to calculate a D statistic (also known as an ABBA-BABA test) to test if *F.* aff. *decumbens* populations are hybrids of *F. californicum* and *F. decumbens*. We used all combinations of *F. californicum*, *F. decumbens*, and *F. mexicanum* as our focal taxa, with *F.* aff. *decumbens* as the potentially hybridized taxon, and ran the test using each focal taxon as the outgroup to polarize SNPs as either ancestral (A) or derived (B). Because individuals from our *F. californicum* sites (Knox and 158) do not form a clade, we calculated D statistics for each collection site separately.

Morphological Analyses

Morphological measurements. We measured ten morphological characters in the field (Table 1) on 68 individuals from 15 sites in Nevada (n = 11), Yuba (n = 19), El Dorado (n = 23), Tehama (n = 4), Mariposa (n = 3), Tulare (n = 1), and Kern (n = 7) counties. Twenty-three were identified as *F. decumbens*, fifteen identified as *F. californicum* and thirty identified as *F.* aff. *decumbens*. For all characters except for plant height and plant width, we made

Downloaded From: https://bioone.org/journals/Madroño on 17 Feb 2026
Terms of Use: https://bioone.org/terms-of-use

five measurements and calculated the mean. Plant height was measured once from the ground to the top of the shrub, and plant width was the widest point. We estimated sepal color and sepal base color as a continuous variable between one and five, with one being yellow and five being orange. We also calculated the range for trichome ray length.

Morphological PCA. We conducted two PCAs of the morphological data to ascertain whether the individuals group together in morphological space and whether these correspond to their morphological identification. For the first PCA, we retained all measured characters and dropped individuals from the analysis that did not have a measurement for every morphological character, because PCA does not allow there to be any missing data. This resulted in 51 individuals being retained. For the second PCA, we dropped measurements that required flowers (three measurements) and again retained only individuals with all measurements present. This resulted in 65 individuals being retained. We then used the R function prcomp from the stats package to calculate principal components and visualized the results using the R function autoplot from the ggplot2 package for both datasets (Wickham 2016).

ANOVA tests. We conducted ANOVA on each measured character separately to determine whether the means were significantly different between our two focal species and *F*. aff. *decumbens*. We conducted analyses using the R packages lme4 and emmeans and used an alpha value of 0.05 (Rv. 4.2.1, R, R Core Team, R foundation for Statistical Computing, Vienna, Austria). We used linear mixed effects models using REML (restricted maximum likelihood) method for each of the plant traits. Species (the two species and the third group of *F*. aff. *decumbens*) was specified as a fixed effect. Species nested within site was a specified as a random effect. For traits where multiple flowers or leaves were sampled per individual plant (e.g., sepal color, petiole length, leaf width, etc.), plant ID was nested within species and site as another random effect. Ray length and plant width to height ratio were log transformed to improve equality of variance. The plant width to height ratio was run with no random effects, and the leaf lobes models were run with only plant ID as a random effect to avoid singularity in the outputs. Contrasts were performed using emmeans to test for significant differences between pairs of species for each plant character/trait. Our sample size was between 51 and 65 for each character, and we did not apply any corrections for multiple testing. Diagnostic plots for each model are available in Appendix S2 Fig. 3.

RESULTS

Molecular Analyses

SNP dataset. After initial processing the final dataset contained 45,871,910 SNPs with 8% missing data and 99 individuals. The data set contained 22 individuals of *F*. *decumbens*, 21 individuals of *F*. *californicum* and 56 potential hybrid individuals (*F*. aff. *decumbens*). The mean depth of coverage per individual was 6.6×. Additional filtering thresholds had minimal effect on estimates of population structure, so all final analyses were performed on the full data set.

Population genomics statistics. Calculated inbreeding coefficients across all individuals ranged between −0.12 and 0.73. All taxa had similar inbreeding coefficients, with *F*. aff. *decumbens* populations being slightly higher (F = 0.264, SE 0.032), followed by *F*. *decumbens* (F = 0.236, SE 0.023) and *F*. *californicum* (F = 0.229, SE 0.025). Inbreeding coefficients for collection site ranged from 0.078–0.623 with site Bennet having the lowest and Dobbins the highest. Boxplots showing the full range of variation across all taxa and collection sites can be found in Appendix S2 Fig. 4. *Fremontodendron decumbens* had the lowest nucleotide diversity (π = 0.00668, SE 0.000026), and *F*. *californicum* and *F*. aff. *decumbens* had similar values (π = 0.00701, SE 0.000026 and π = 0.00738, SE 0.000027). When we calculated nucleotide diversity across collection sites, Dobbins had the lowest (π = 0.00438) and McCourtney Road had the highest (π = 0.00708). Boxplots showing the full range of variation across all taxa and collection sites can be found in Appendix S2 Fig. 5. *Fremontodendron decumbens* had the highest pairwise Fst when compared to *F*. *californicum* (0.10), followed by *F*. *decumbens* compared with *F*. aff. *decumbens* (0.075), then *F*. *californicum* by *F*. aff. *decumbens* (0.038). Pairwise Fst by collection site ranged from −0.012 to 0.256, and a plot showing all comparisons can be found in Appendix S2 Figs. 6 and 7.

Genomic principal components analysis. After pruning for linkage disequilibrium, we retained 9,203,312 SNPs and 99 individuals in the dataset. PC axes one and two together explained ~18.81% of the total genomic variation. PCA analysis showed clear separation between two groups on PC1, all of *F*. *decumbens* in one group and all of *F*. *californicum* and *F*. aff. *decumbens* in the second group. PC2 separates *F*. *californicum* and *F*. aff. *decumbens* samples into three groups, none of which correspond to morphological species assignments (Fig. 3). There is some degree of geographic structuring evident; collection sites cluster together, and samples from Napa, Shasta, and Tehama counties group together, while the other clusters are samples from Nevada and Yuba counties and one just representing Nevada County (Appendix S2 Fig. 8).

Admixture. When we ran an unsupervised population genomic structure analysis assuming two ancestral populations (K = 2, theoretically corresponding to *F*. *californicum* and *F*. *decumbens*) on the LD pruned dataset, all *F*. *californicum* individuals were assigned to one ancestral population and all *F*. *decumbens*

Downloaded From: https://bioone.org/journals/Madroño on 17 Feb 2026
Terms of Use: https://bioone.org/terms-of-use

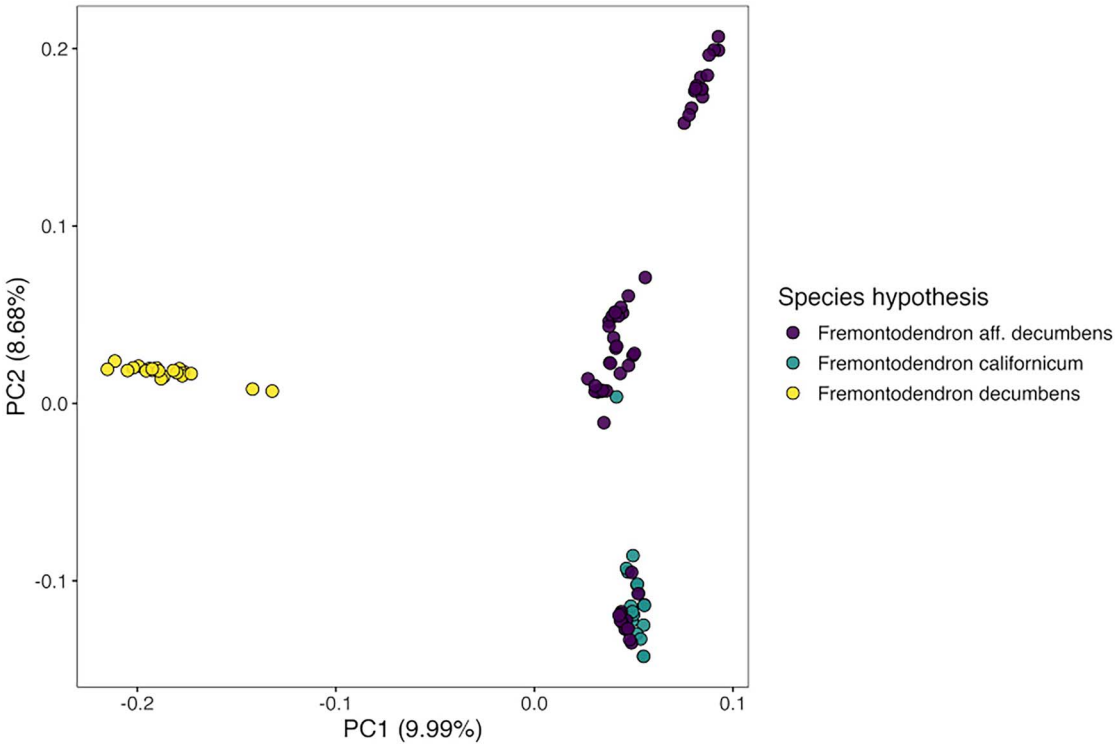

FIG. 3. Principle components analysis of genomic variation in the sampled *Fremontodendron* individuals. Each point corresponds to one individual, and points are colored by the assigned species hypothesis based on morphology.

individuals to the other. Twenty-two *F.* aff. *decumbens* individuals were assigned mixed ancestry from *F. decumbens* and *F. californicum,* and the rest were assigned 100% *F. decumbens* ancestry (Fig. 4). When we ran a supervised analysis, where we fixed the percent identity of individuals identified morphologically as *F. decumbens* or *F. californicum* at 100% from their respective species, and only estimated the ancestral proportions for samples identified as *F.* aff. *decumbens*, all *F.* aff. *decumbens* were assigned essentially 100% *F. californicum* ancestry (Fig. 4).

When we ran admixture using K = 3, *F. decumbens* were all assigned 100% ancestry to one group, and *F. californicum* and *F.* aff. *decumbens* individuals were split between the second and third groups. The second group mostly consisted of northern populations of *F. californicum* and *F.* aff. *decumbens* from the Hwy 158, and Whiskeytown sites but also included individuals from the Bennet site. More southerly *F. californicum* and *F.* aff. *decumbens* (Knox and Dobbins sites) were assigned 100% identity to the third group, and the rest of the *F.* aff. *decumbens* sites were admixed between the second and third groups (Fig. 4). When we ran admixture using K = 4, we see similar groups as above, except that the Bennet and Knox sites grouped together, and Brownsville, Dobbins, and Frenchtown Road formed the fourth group, leaving only McCourtney as admixed (Fig. 4) Cross validation analyses run on models assuming up to five ancestral populations indicated two ancestral populations (K = 2) to be the mostly likely scenario (Appendix S2 Fig. 9).

F3 statistics. After removing sites with missing data and pruning for linkage disequilibrium, we retained 1,729,585 SNPs and 99 individuals in the dataset. F3 statistics calculated where populations were assigned to morphological species hypotheses were not negative for any of the three possible topologies, indicating that the populations can be fitted to a tree, providing no evidence of hybridization. F3 statistics calculated where populations were assigned to geographic groups were only negative when the McCourtney population was tested as a hybrid of Bennet and Knox populations, indicating that the triplet containing McCourtney, Bennet, and Knox populations cannot be fit to a tree (Fig. 5, Appendix S2 Fig. 10). As none of these populations represent known *F. decumbens*, this does not support the hypothesis that any of the potential hybrid populations are in fact hybrids of *F. decumbens* and *F. californicum*.

Treemix. Our input dataset contained 1,292,349 SNPs and 108 individuals. When we ran Treemix with individuals pooled by collection site, our OptM analysis indicated that three migration edges were optimal across all tested roots. With three migration

Downloaded From: https://bioone.org/journals/Madroño on 17 Feb 2026
Terms of Use: https://bioone.org/terms-of-use

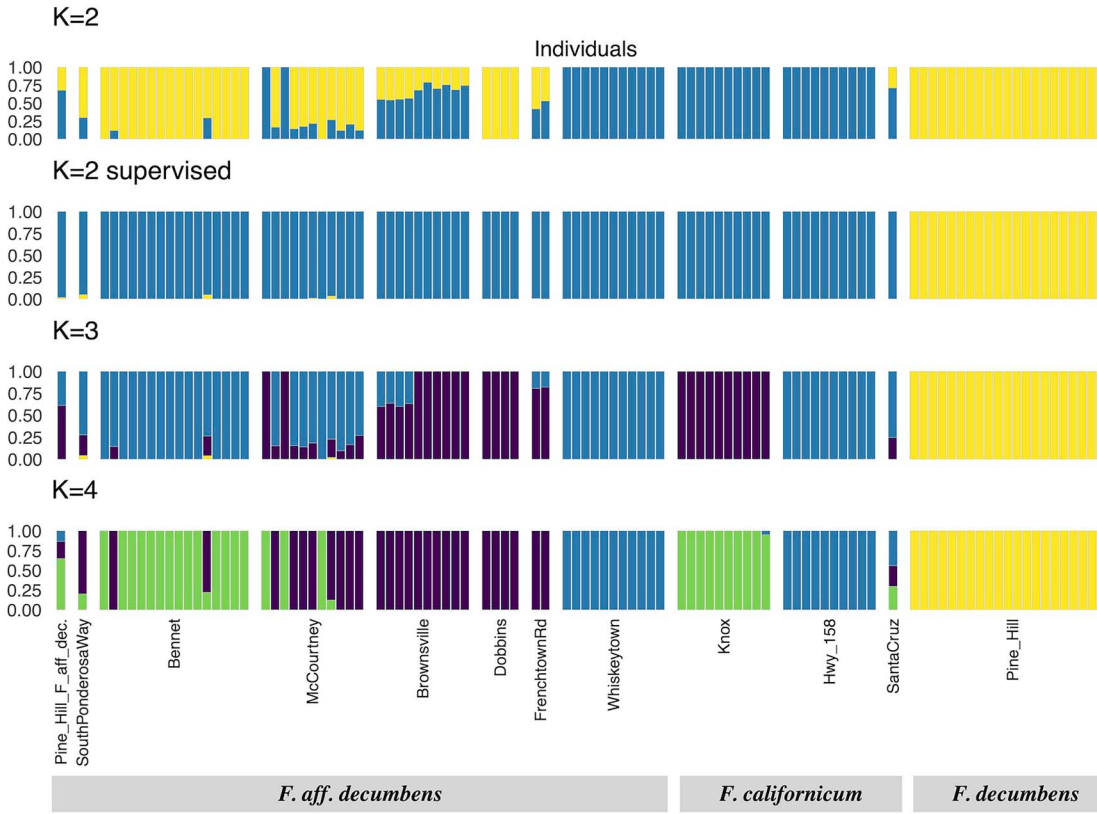

FIG. 4. Ancestry proportions generated using ADMIXTURE v. 1.3. Each bar represents one individual, the color of the bar represents the proportion of predicted ancestry, and the bars are grouped according to collection site and species hypothesis based off of morphology. K = 2 supervised, indicates that for this analysis the ancestry of *F. californicum* (Knox and 158 sites) was assumed to be 100% from one group and the ancestry of *F. decumbens* (Pine Hill site) was assumed to be 100% from the other group. Then ancestry fractions were estimated for all other samples.

edges, admixture edges are consistently drawn between Knox and McCourtney collection sites, Knox and the common ancestor of 158 and Whiskeytown sites, and Knox and *F. mexicanum* (Fig. 6). This last admixture edge seems likely to represent admixture between the Knox site and more southerly *F. californicum* populations, which may be more genetically similar to *F. mexicanum*.

Maximum likelihood phylogeny. The final data matrix we used to construct the maximum likelihood tree contained 111 individuals, 1,367,652 sites, and 357,415 parsimony informative sites. Clades in the tree largely conformed with groups observed in the PCA and Admixture analyses and all branches had high support values (Fig. 7). Notably, *F. decumbens* individuals formed a clade with individuals from Whiskeytown and 158 sites, whereas in the Treemix trees, *F. decumbens* was often placed near *F. mexicanum*.

D statistics. Our final dataset contained 111 individuals and 1,367,652 SNPs. This represents the same 99 individuals as the other analyses plus the same 12 *F. mexicanum* individuals as above. Across the four different root topologies tested, our D statistics indicated introgression between *F.* aff. *decumbens* and *F. decumbens* 12 times, *F.* aff. *decumbens* and *F. californicum* nine times, and *F.* aff. *decumbens* and *F. mexicanum* eight times (Table 1).

Morphological Analyses

Morphological PCA. PC axis 1 of the PCA that included all traits measured explained ~40.75% of the total morphological variance and separated *F. decumbens* from *F.* aff. *decumbens* and *F. californicum*. *F.* aff. *decumbens* and *F. californicum* have largely overlapping 95% confidence ellipses, although the 95% confidence ellipse for *F.* aff. *decumbens* is slightly shifted towards *F. decumbens*. PC axis 2 of the PCA that included all traits measured explained ~19.1% of the total morphological variance and did not separate any groups (Fig. 8). The PCA analysis where floral traits were dropped, did not separate any groups.

F tests. *Fremontodendron*. aff. *decumbens* individuals grouped with *F. californicum* for most traits used in species identification, such as sepal color, plant

Downloaded From: https://bioone.org/journals/Madroño on 17 Feb 2026
Terms of Use: https://bioone.org/terms-of-use

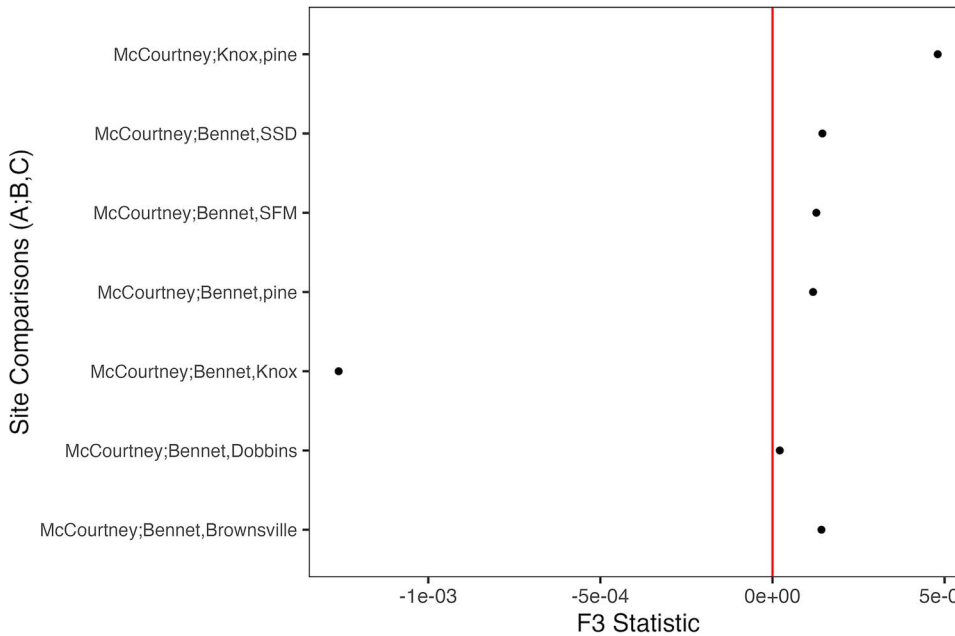

FIG. 5. The seven lowest f3 statistics. Y axis shows site comparisons in the format (A;B, C), where A is being tested as a hybrid of B and C. The vertical line is placed at x = 0. F3 statistics less than zero can be interpreted as evidence of hybridization. A plot of all f3 statistics can be found in Appendix S2 Fig. 10.

height-width ratio, and trichome length. They grouped with *F. decumbens* for only one trait, plant height (Table 2). Plots of raw morphological data can be found in Appendix S2 Fig. 11.

DISCUSSION

Our results show that plants found on gabbro soils at Pine Hill (including the adjacent Fairview Road area) in El Dorado County that are unambiguously identified as *F. decumbens* based on morphology form a well-supported, coherent genetic group. In contrast, *F.* aff. *decumbens* individuals formed multiple genetic groups, often clustering with *F. californicum*. Indeed, our *F.* aff. *decumbens* individuals came from two geographically and genetically distinct groups of plants that have largely similar morphologies. This could either be a case of completely independent evolution of this phenotype, or, more likely, indicative that there is a large degree of standing variation in *F. californicum* and that these two groups inhabit a similar selective environment. On our phylogenies, the *F.* aff. *decumbens* individuals tend to form clades with the geographically closest *F. californicum*. The phylogenetic position of *F. decumbens* is somewhat ambiguous, but all of the *F. decumbens* individuals form a clade that received high bootstrap support.

We found little evidence to suggest that the individuals designated as *F.* aff. *decumbens* are in fact derived from recent or ancestral hybridization events between *F. decumbens* and *F. californicum*. Rather, our data indicate that *F.* aff. *decumbens* individuals fall within the scope of variation encompassed by *F. californicum*. It is well-established that *F. californicum* is a highly variable species. Our data suggest that a decumbent habit and/or somewhat darker flowers, as found in *F.* aff. *decumbens*, do not indicate hybrid parentage. Most of our analyses show no evidence of ancestral or modern gene flow between *F. decumbens* and *F.* aff. *decumbens*. In fact, these analyses instead indicate that there has been gene flow between populations identified unambiguously as *F. californicum* and those provisionally designated as *F.* aff. *decumbens* but not between the latter and *F. decumbens*. Only two of our six analyses indicated that the *F.* aff. *decumbens* individuals could possibly have *F. decumbens* ancestry, but we question the validity of those results as explained below.

One was our unsupervised ADMIXTURE K = 2 analysis. When we compare the results of the K = 2 plots and the K = 3 plots, we can see that they both assigned mixed ancestry to the McCourtney and Brownsville (*F.* aff. *decumbens*) collection sites (Fig. 4). However, the supervised model, where known *F. californicum* and known *F. decumbens* are fixed as two separate populations and ancestry was only estimated for the remaining (*F.* aff. *decumbens*) individuals, assigned all *F.* aff. *decumbens* individuals to the same population as *F. californicum*. Because the unsupervised K = 2 analysis grouped *F. decumbens* and the Bennet site (*F.* aff. *decumbens*) together and assigned admixed ancestry to the McCourtney and Brownsville collections sites, it partitioned the

Downloaded From: https://bioone.org/journals/Madroño on 17 Feb 2026
Terms of Use: https://bioone.org/terms-of-use

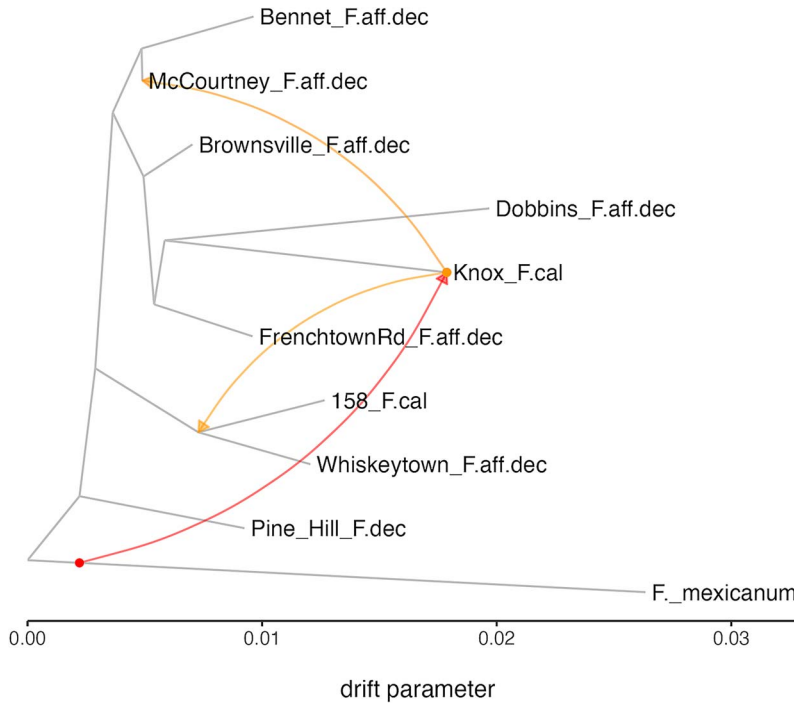

FIG. 6. Admixture graph generated using Treemix and our VCF file that includes 12 *F. mexicanum* individuals. Three migration edges (orange and red arrows) were selected as the optimal number to add using the R package OptM. This tree is rooted on *F. mexicanum* for visualization purposes, but we examined trees rooted on all species aside from *F.* aff. *decumbens*.

ancestry of the McCourtney and Brownsville sites between the two groups. This resulted in partial *F. decumbens* ancestry for the McCourtney and Brownsville sites and 100% *F. decumbens* ancestry for the Bennet site. When we defined *F. decumbens* as its own group, we saw unequivocal grouping of *F.* aff. *decumbens* with *F. californicum*. This indicates that the partial *F. decumbens* ancestry assigned to the McCourtney and Brownsville sites by the unsupervised K = 2 model was caused by the grouping of the Bennet site with *F. decumbens*. This is supported by the fact that the K = 3 analysis assigned *F. decumbens* to its own group, as in the K = 2 supervised analysis. This then caused the ancestry of the McCourtney and Brownsville collection sites to be partitioned between the two *F. californicum* groups. Because of the pattern in the K = 3 analysis, the fact that fixing the ancestries of some populations in ADMIXTURE (supervised analysis) should decrease the error in the models, and the evidence from our other analyses, we conclude that the pattern represented by the K = 2 supervised and K = 3 analyses is most trustworthy.

The other analysis that indicated some possibility of ancestral hybridization events between *F. decumbens* and *F.* aff. *decumbens* was our D statistic analysis; however, the same analysis indicated hybridization between *F. californicum*, *F. mexicanum*, and *F. decumbens* as well (Table 2). The analysis returned conflicting results depending on which individuals we used to represent *F. californicum,* as well as which taxon we used as the outgroup. D statistics are known to be vulnerable to false positive results for a variety of reasons, including the presence of unsampled lineages, ancestral population structure, and hybridization involving closely related populations that are not being tested (Theunert and Slatkin 2017; Tricou et al. 2022). The fact that we saw different results depending on which individuals we included in the analysis confounds the interpretation and suggests that these results are not reflective of true hybridization events. One factor that may be influencing the results is that *F. mexicanum* may have been involved in ancestral hybridization events with *F. californicum* that are skewing the results. This is hinted at by our Treemix tree, which draws an admixture edge between the base of the *F. mexicanum* branch and the Knox site, as well as the fact that eight of the D statistics we calculated indicate introgression between *F.* aff. *decumbens* and *F. mexicanum* (Fig. 6, Table 2). This possibility is also supported by unpublished data from our lab that shows significant admixture proportions from *F. mexicanum* in some *F. californicum* individuals from southern California. Because the tests returned conflicting results, it is difficult to assign any interpretation to this outcome.

In contrast to the inconclusive results discussed above, our Treemix, f3, PCA, and the majority of

Downloaded From: https://bioone.org/journals/Madroño on 17 Feb 2026
Terms of Use: https://bioone.org/terms-of-use

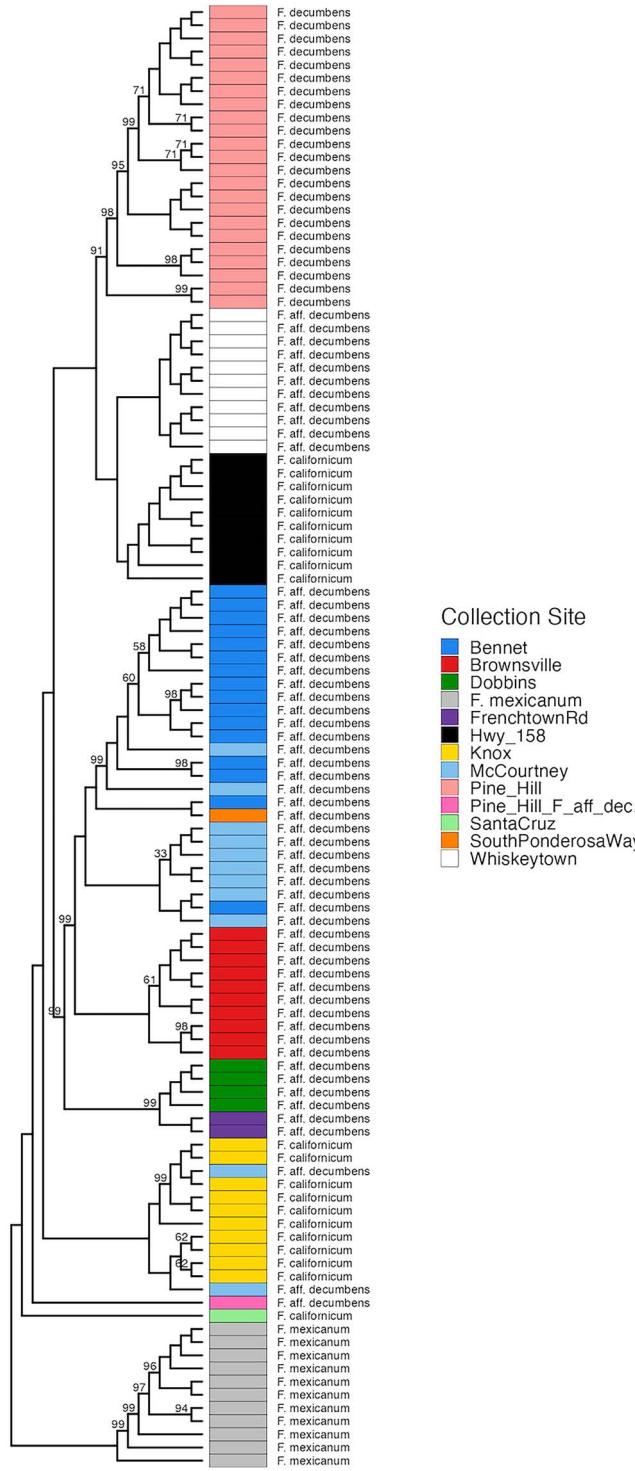

FIG. 7. Maximum likelihood tree constructed using IQ-TREE. Tips represent individual samples, branches that are not fully supported display branch support values calculated from 1000 ultrafast bootstraps in IQ-TREE, and colored boxes indicate geographic populations.

Downloaded From: https://bioone.org/journals/Madroño on 17 Feb 2026
Terms of Use: https://bioone.org/terms-of-use

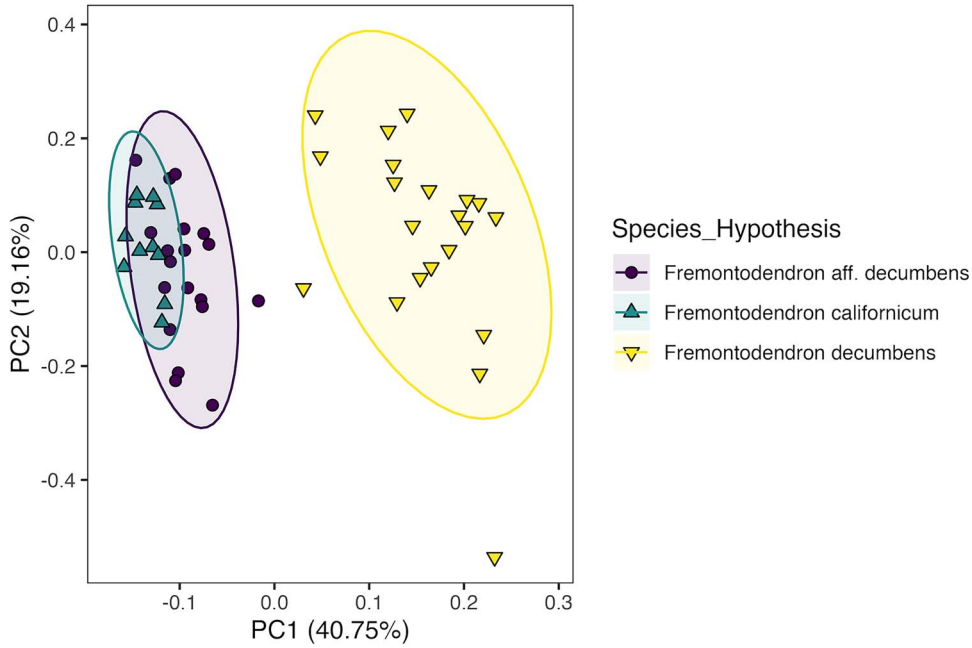

FIG. 8. Principal components analysis of all morphological traits measured on *Fremontodendron* individuals. Each dot represents one individual, and the ellipses represent the 95% confidence interval around each group.

our ADMIXTURE runs show no evidence of ancestral or modern gene flow between *F. decumbens* and *F.* aff. *decumbens*. In fact, these analyses indicate that there has been gene flow between the *F. californicum* individuals from the Knox site and *F.* aff. *decumbens* individuals from the McCourtney site. It is important to note that our methods do not test for introgression at deeper phylogenetic time scales. However, the close degree of relatedness between *F. decumbens* and *F. californicum* makes it plausible that population genetic approaches would be able to detect introgression. We conclude that there is little support for the hypothesis that the *F.* aff. *decumbens* individuals are the result of introgression between *F. californicum* and *F. decumbens* and significant evidence that the *F.* aff. *decumbens* individuals are genetically more similar to *F. californicum* individuals.

Our results showing that *F. decumbens* represents a genetically cohesive and distinct group from *F. californicum* are consistent with those of Kelman et al. (2006), while our rejection of the hybrid origin hypothesis for the *F.* aff. *decumbens* individuals is somewhat contrary to their results. We note, however, that Kelman et al.'s (2006) study was conducted on 37 individuals genotyped for 66 AFLPs, which are dominant markers, while the current study encompassed 99 individuals collected from a larger geographic area and genotyped for 45,871,910 SNPs (Kelman et al. 2006).

Taxonomically, it seems clear that the *F.* aff. *decumbens* individuals should be included in *F.*

TABLE 2. MORPHOLOGICAL TRAITS MEASURED IN THE FIELD. 1, 2, or 3 denotes the groups output from an emmeans contrasts test and Prob > F indicates statistical significance. Values marked with a * are significant.

Trait	*F. californicum*	*F.* aff. *decumbens*	*F. decumbens*	Prob > F
Width	1	1	1	0.1573
Height	1	1	2	0.002009*
Width–Height Ratio	1	1	2	<0.0001*
Sepal Color	1	1	2	<0.0001*
Sepal Base Color	2	1	3	<0.0001*
Peduncle Length	1	1	2	0.000716*
Leaf Length	1	1	1	0.8174
Petiole Length	1	1	1	0.5267
Leaf Width	1	1	1	0.8426
Leaf Lobes	1	1	1	0.4095
Ray Length	1	1	1	0.1573

Downloaded From: https://bioone.org/journals/Madroño on 17 Feb 2026
Terms of Use: https://bioone.org/terms-of-use

californicum. Prior work has established that there are clear morphological characters separating *F. decumbens* and *F. californicum* (Kelman 1991) and our results also indicate that consistently darker flowers, a more decumbent habit, and longer trichome rays distinguish the former from the latter. Somewhat intermediate plants such as the *F.* aff. *decumbens* individuals can be distinguished from *F. decumbens* in that they do not have both the decumbent habit and dark sepals that are the most diagnostic traits of *F. decumbens*. Our morphological data indicates that *F.* aff. *decumbens* individuals typically are similar in height to *F. decumbens,* but much less wide, suggesting that they do not truly have a decumbent stature. Additionally, they have sepals that are somewhat darker than *F. californicum,* but not as dark as *F. decumbens*. Some keys may need to be updated to allow for the occasional occurrence of *F. decumbens*-like characters in otherwise *F. californicum* individuals. Although our maximum likelihood tree showed *F. decumbens* to be nested within *F. californicum*, our Treemix tree placed it as the outgroup to the clade containing the *F. californicum* and *F.* aff. *decumbens* individuals. Even if *F. decumbens* is truly nested within *F. californicum*, we support its recognition at the species level. Our results as well as others show that it is morphologically and genetically distinct from *F. californicum*. Additionally, speciation caused by colonizing a new soil environment can often result in a lack of reciprocal monophyly, especially if the lineage separation is relatively recent (Baldwin 2005; Grossenbacher et al. 2014; Rajakaruna 2018).

The current recovery plan for *F. decumbens* states that decumbent plants in Yuba and Nevada counties should be secured and protected until determined not to be *F. decumbens* (USFWS 2002). Our study has not provided any evidence to suggest that these plants are genetically aligned with *F. decumbens*, and we find no reason to believe that further sampling would result in a different outcome. However, the high rates of inbreeding found among most populations of *F. californicum* suggest that many of these populations could be uniquely adapted to their local environments and still constitute an important aspect of the gabbro flora of the Sierra Nevada foothills, worthy of conservation.

Data Accessibility

Raw sequence data can be downloaded from GenBank at BioProject (PRJNA1041826). Our SNP and morphological datasets can be downloaded from Data Dryad (10.5061/dryad.xsj3tx9s2). Scripts used for data analysis can found on Github (https://github.com/reedjohnkenny/Fremontodendron).

Acknowledgments

We would like to thank the Bureau of Land Management, the California Conservation Genomics Project, and the El Dorado and Redbud chapters of the California Native Plant Society for funding this work. Support from the UC Davis Agricultural Experiment Station (Project Number CA-D-PLS-6273-H) is also gratefully acknowledged. We thank Bruce Rannala and Jeff Ross-Ibarra for their comments on the manuscript and Peter Lash for assisting with field collections. We thank the UC Davis herbarium for housing specimens collected from this project. We additionally thank two anonymous reviewers for comments that improved the manuscript.

Literature Cited

Alexander, D. H. and K. Lange. 2011. Enhancements to the ADMIXTURE algorithm for individual ancestry estimation. BMC Bioinformatics 12:246.

———, J. Novembre, and K. Lange. 2009. Fast model-based estimation of ancestry in unrelated individuals. Genome Research 19:1655–1664.

Auwera, G. V. der, and B. D. O'Connor. 2020. Genomics in the cloud: using Docker, GATK, and WDL in Terra. First edition. O'Reilly, Beijing Boston Farnham Sebastopol, Tokyo.

Baldwin, B. G. 2005. Origin of the serpentine-endemic herb *Layia discoidea* from the widespread *L. glandulosa* (Compositae). Evolution 59:2473–2479.

Chen, S., Y. Zhou, Y. Chen, and J. Gu. 2018. fastp: an ultra-fast all-in-one FASTQ preprocessor. Bioinformatics 34:i884–i890.

Danecek, P., A. Auton, G. Abecasis, C. A. Albers, E. Banks, M. A. DePristo, R. E. Handsaker, et al. 2011. The variant call format and VCFtools. Bioinformatics 27:2156–2158.

Fitak, R. R. 2021. OptM: estimating the optimal number of migration edges on population trees using Treemix. Biology Methods and Protocols 6:bpab017.

Gross, B. L. and L. H. Rieseberg. 2005. The ecological genetics of homoploid hybrid speciation. Journal of Heredity 96:241–252.

Grossenbacher, D. L., S. D. Veloz, and J. P. Sexton. 2014. Niche and range size patterns suggest that speciation begins in small, ecologically diverged populations in North American monkeyflowers (Mimulus spp.). Evolution 68:1270–1280.

Hoang, D. T., O. Chernomor, A. von Haeseler, B. Q. Minh, and L. S. Vinh. 2018. UFBoot2: improving the ultrafast bootstrap approximation. Molecular Biology and Evolution 35:518–522.

Hubisz, M. J., D. Falush, M. Stephens, and J. K. Pritchard. 2009. Inferring weak population structure with the assistance of sample group information. Molecular Ecology Resources 9:1322–1332.

Kelman, W., L. Broadhurst, C. Brubaker, and A. Franklin. 2006. Genetic relationships among Fremontodendron (Sterculiaceae) populations of the central Sierra Nevada foothills of California. Madroño 53:380–387.

Kelman, W. M. 1991. A revision of Fremontodendron (Sterculiaceae). Systematic Botany 16:3–20.

Korunes, K. L. and K. Samuk. 2021. pixy: unbiased estimation of nucleotide diversity and divergence in the presence of missing data. Molecular Ecology Resources 21:1359–1368.

Li, H. and R. Durbin. 2009. Fast and accurate short read alignment with Burrows–Wheeler transform. Bioinformatics 25:1754–1760.

Linck, E. and C. J. Battey. 2019. Minor allele frequency thresholds strongly affect population structure inference

Downloaded From: https://bioone.org/journals/Madroño on 17 Feb 2026
Terms of Use: https://bioone.org/terms-of-use

with genomic data sets. Molecular Ecology Resources 19:639–647.

LLOYD, R. M. 1965. A new species of Fremontodendron (Sterculiaceae) from the Sierra Nevada foothills, California. Brittonia 17:382–384.

MEDEIROS, I. D., N. RAJAKARUNA, AND E. B. ALEXANDER. 2015. Gabbro soil-plant relations in the California Floristic Province. Madroño 62:75–87.

MINH, B. Q., H. A. SCHMIDT, O. CHERNOMOR, D. SCHREMPF, M. D. WOODHAMS, A. VON HAESELER, AND R. LANFEAR. 2020. IQ-TREE 2: new models and efficient methods for phylogenetic inference in the genomic era. Molecular Biology and Evolution 37:1530–1534.

MIRCHANDANI, C. D., A. J. SHULTZ, G. W. C. THOMAS, S. J. SMITH, M. BAYLIS, B. ARNOLD, R. CORBETT-DETIG, ET AL. 2024. A fast, reproducible, high-throughput variant calling workflow for population genomics. Molecular Biology and Evolution 41:msad270.

ORTIZ, E. M. 2019. vcf2phylip v2.0: convert a VCF matrix into several matrix formats for phylogenetic analysis.

PATTERSON, N., P. MOORJANI, Y. LUO, S. MALLICK, N. ROHLAND, Y. ZHAN, T. GENSCHORECK, ET AL. 2012. Ancient admixture in human history. Genetics 192:1065–1093.

PETER, B. M. 2022. A geometric relationship of F2, F3 and F4-statistics with principal component analysis. Philosophical Transactions of the Royal Society B 377:20200413.

PETR, M., B. VERNOT, AND J. KELSO. 2019. admixr—R package for reproducible analyses using ADMIXTOOLS. Bioinformatics 35:3194–3195.

PICKRELL, J. K. AND J. K. PRITCHARD. 2012. Inference of population splits and mixtures from genome-wide allele frequency data. PLOS Genetics 8:e1002967.

POPLIN, R., V. RUANO-RUBIO, M. A. DEPRISTO, T. J. FENNELL, M. O. CARNEIRO, G. A. VAN DER AUWERA, D. E. KLING, ET AL. 2017. Scaling accurate genetic variant discovery to tens of thousands of samples. bioRxiv 201178; doi: https://doi.org/10.1101/201178

PURCELL, S., B. NEALE, K. TODD-BROWN, L. THOMAS, M. A. R. FERREIRA, D. BENDER, J. MALLER, ET AL. 2007. PLINK: a tool set for whole-genome association and population-based linkage analyses. American Journal of Human Genetics 81:559–575.

RAJAKARUNA, N. 2018. Lessons on evolution from the study of edaphic specialization. The Botanical Review 84:39–78.

RYAN, M. E., J. R. JOHNSON, AND B. M. FITZPATRICK. 2009. Invasive hybrid tiger salamander genotypes impact native amphibians. Proceedings of the National Academy of Sciences 106:11166–11171.

TARASOV, A., A. J. VILELLA, E. CUPPEN, I. J. NIJMAN, AND P. PRINS. 2015. Sambamba: fast processing of NGS alignment formats. Bioinformatics 31:2032–2034.

THEUNERT, C. AND M. SLATKIN. 2017. Distinguishing recent admixture from ancestral population structure. Genome Biology and Evolution 9:427–437.

TRICOU, T., E. TANNIER, AND D. M. DE VIENNE. 2022. Ghost lineages highly influence the interpretation of introgression tests. Systematic Biology 71:1147–1158.

USFWS. 1996. Endangered and threatened wildlife and plants; Determination of endangered status of four plants and threatened status of one plant from the central Sierran foothills of California. United States Fish and Wildlife Service. Washington, D.C. Website https://www.federalregister.gov/documents/1996/10/18/96-26740/endangered-and-threatened-wildlife-and-plants-determination-of-endangered-status-for-four-plants-and [accessed 04 May 2025].

USFWS. 2002. Recovery plan for gabbro soil plants of the central Sierra Nevada foothills. United States Fish and Wildlife Service. Washington, D.C. Website https://ecos.fws.gov/docs/recovery_plan/020830b.pdf [accessed 04 May 2025].

WICKHAM, H. 2016. ggplot2: elegant graphics for data analysis, 2nd ed. Springer International Publishing: Imprint: Springer, Cham.

YU, G. 2020. Using ggtree to visualize data on tree-like structures. Current Protocols in Bioinformatics 69:e96.

Downloaded From: https://bioone.org/journals/Madroño on 17 Feb 2026
Terms of Use: https://bioone.org/terms-of-use